

A. Contributions

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Implementation & Iteration:

- *Object Head:* Bowen Yang, Xian Nie, Cunxin Fan, Xiaosong Jia.
- *Depth Head:* Xian Nie, Yilin Chai, Zijian Liang, Bowen Yang, Cunxin Fan, Xiaosong Jia.
- *Skill Head:* Zuhao Ge, Xian Nie, Bowen Yang, Cunxin Fan, Xiaosong Jia.
- *Head Merge:* Bowen Yang, Xian Nie, Zuhao Ge, Cunxin Fan, Xiaosong Jia.
- *Codebase & Infra:* Bowen Yang, Cunxin Fan, Xiaosong Jia.

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Real-Robot Experiments: Zuhao Ge, Yuchen Zhou, Yufeng Li, Chao Jing.

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B. Complete Results for All Datasets

For Real robot experiments, we provide the complete results across all 6 tasks in Table III.

For LIBERO-plus [28] benchmark, we provide the complete results in Table IV.

For the RoboTwin 2.0 [16] benchmark, we provide the complete results in Table V.

C. Implementation Details of Each Head and Adapter

Object Head. We implement object-level attention supervision by maximizing the attention mass assigned to stage-specific object masks, matching Eq. 4 in the main paper. Algorithm 2 outlines the procedure for computing the object grounding loss. It selects a subset of attention heads, indexes image-view patches, converts stage-aware masks into valid object regions, and penalizes attention mass outside the target region. The supervision is applied across multiple transformer layers and averaged to produce the final loss.

Depth Head. To integrate depth information into cross-modal attention, we use a specialized Key-Value (KV) projector that maps depth tokens into compatible representations for selected attention heads. Algorithm 3 provides the implementation of the DepthKVProjector and how it is used to modify attention computation. Standard heads use key and value states from VLM backbone, while selected heads attend to projected depth tokens, supporting geometry-aware reasoning.

Skill Head. The Skill Head encourages semantic grounding by matching attention-derived features to a soft skill distribution target. Algorithm 4 describes the KL loss computation

pipeline. For each transformer layer, we extract the action-query attention output, average features, and apply a classification head. The output is compared to normalized histogram targets, capturing the distribution of skill labels across the trajectory.

ControlNet-style Adapter. To enable fine-grained control signal injection, we design a ControlNet-inspired dual-path attention mechanism. Algorithm 5 shows the implementation of ControlAttention, which splits the attention computation into a main path and a control-specific branch. The outputs from both branches are fused using a zero-initialized linear projection, allowing conditional modulation without disrupting pretrained behavior.

D. Implementation Details of Factor Quality Ablation

This part provides detailed explanations corresponding to Section III-C and Section V-B of the main paper.

Object Head. To assess the impact of factor quality, we require precise control over the model’s grounding strength. While the standard supervision in Eq. 4 of the main paper encourages the model to maximize attention on the object, this ablation study necessitates clamping the intensity to specific scalar values $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$. We define the grounding strength m as the cumulative attention mass falling within the ground-truth region mask $\mathcal{M}$:

$$m = \sum_{j \in \mathcal{M}} A_j, \quad (10)$$

where A_j are the attention weights of the action token. Since the total attention is normalized, m directly represents the concentration of focus on the target object. For this experiment, we replace the standard objective with a regression loss to force m towards the target α :

$$\mathcal{L}_{\text{ablation}} = \begin{cases} \frac{0.5(m - \alpha)^2}{\beta}, & \text{if } |m - \alpha| < \beta, \\ |m - \alpha| - 0.5\beta, & \text{otherwise.} \end{cases} \quad (11)$$

where the smoothing parameter β is set to 0.05. This objective allows us to strictly regulate the grounding quality for sensitivity analysis.

For the baseline model π_0 (which lacks explicit object supervision), we measure its intrinsic grounding capability using the same metric m . The reported value (26.5%) is obtained by averaging m over 200 evaluation steps during inference. This result indicates that, in the absence of targeted supervision, the model exhibits a natural tendency to attend to task-irrelevant objects, thereby motivating our design choice to introduce auxiliary guidance.

Depth Head. Unlike the Object and Skill heads, the Depth head is enforced via architectural constraints (attention injection) rather than optimization objectives. Therefore, we cannot regulate its quality through loss scaling. Instead, we control the strength of geometric cues by modulating the signal-to-noise ratio of the input features. Let $\mathbf{f}_{\text{depth}}$ denote the feature extracted from the frozen depth encoder, and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ be a Gaussian noise vector normalized to match the statistics of